Voevodsky's Lectures on Motivic Cohomology 2000/2001

Pierre Deligne

Abstract These lectures cover several important topics of motivic homotopy theory which have not been covered elsewhere. These topics include the definition of equivariant motivic homotopy categories, the definition and basic properties of solid and ind-solid sheaves and the proof of the basic properties of the operations of twisted powers and group quotients relative to the $\mathbf{A}^1$-equivalences between ind-solid sheaves.

1 Introduction

The lectures which provided the source for these notes covered several different topics which are related to each other but which do not in any reasonable sense form a coherent whole. As a result, this text is a collection of four parts which refer to each other but otherwise are independent.

In the first part we introduce the motivic homotopy category and connect it with the motivic cohomology theory discussed in [7]. The exposition is a little unusual because we wanted to avoid any references to model structures and still prove the main theorem 2. We were able to do it modulo 6 where we had to refer to the next part.

The second part is about we the motivic homotopy category of G-schemes where G is a finite flat group scheme with respect to an equivariant analog of the Nisnevich topology. Our main result is a description of the class of $\mathbf{A}^1$-equivalences (formerly called $\mathbf{A}^1$-weak equivalences) given in Theorem 4 (also in Theorem 5). For the trivial group G we get a new description of the $\mathbf{A}^1$-equivalences in the non equivariant setting.

In the third part we define a class of sheaves on G-schemes which we call solid sheaves. It contains all representable sheaves and quotients of representable sheaves by subsheaves corresponding to open subschemes. In particular the Thom

P. Deligne
School of Mathematics, Institute for Advanced Study, Princeton, NJ 08540, USA
e-mail: deligne@math.ias.edu

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spaces of vector bundles are solid sheaves. The key property of solid sheaves can be expressed by saying that any right exact functor which takes open embeddings to monomorphisms is left exact on solid sheaves. A more precise statement is Theorem 6.

In the fourth part we study two functors. One is the extension to pointed sheaves of the functor from G-schemes to schemes which takes X to X/G. The other one is extension to pointed sheaves of the functor which takes X to X^W where W is a finite flat G-scheme. We show that both functors take solid sheaves to solid sheaves and preserve local and $\mathbf{A}^1$-equivalences between termwise (ind-)solid sheaves.

The material of all the parts of these notes but the first one was originally developed with one particular goal in mind – to extend non-additive functors, such as the symmetric product, from schemes to the motivic homotopy category. More precisely, we were interested in functors given by

$$T : X \mapsto (X^W \times E)/G$$

where G is a finite flat group scheme, W is a finite flat G-scheme and E any G-scheme of finite type. The equivariant motivic homotopy category was introduced to represent T as a composition

$$X \mapsto X^W \mapsto X^W \times E \mapsto (X^W \times E)/G$$

and solid sheaves as a natural class of sheaves on which the derived functor $\mathbf{L}T$ coincides with T.

In the present form these notes are the result of an interactive process which involved all listeners of the lectures. A very special role was played by Pierre Deligne. The text as it is now was completely written by him. He also cleared up a lot of messy parts and simplified the arguments in several important places.

Princeton, NJ, 2001.
Vladimir Voevodsky

2 Motivic Cohomology and Motivic Homotopy Category

We will recall first some of last year results (see [7]).

2.1 Last Year

1.1 We work over a field k which sometimes will have to be assumed to be perfect. The schemes over k we consider will usually be assumes separated and smooth of finite type over k. We note Sm/k their category. Three Grothendieck topologies on Sm/k will be useful: Zariski, Nisnevich and etale. For each of these topologies a sheaf on (Sm/k) amounts to the data for X smooth over k of a sheaf F_X on the small site X_{Zar} (resp. X_{Nis}, X_{et}) of the open subsets U of X (resp. of $U \to X$ etale), with F_X functorial in X: a map $f : X \to Y$ induces $f^* : f^* F_Y \to F_X$.

1.2 The definition of the motivic cohomology groups of X smooth over k has the following form:

(a) One defines for each $q \in \mathbf{Z}$ a complex of presheaves of abelian groups $\mathbf{Z}(q)$ on Sm/k. It is in fact a complex of sheaves for the etale topology, hence a fortiori for the Nisnevich and Zariski topology. For any abelian group A the same applies to $A(q) := A \otimes \mathbf{Z}(q)$.

(b) The motivic cohomology groups of X with coefficients in A are the hypercohomology groups of the $A(q)$, in the Nisnevich topology:

$$H^{p,q}(X, A) := \mathbf{H}^p(X_{Nis}, A(q))$$

For $A = \mathbf{Z}$ we will write simply $H^{p,q}(X)$.

Motivic cohomology has the following properties:

1. The complex $\mathbf{Z}(q)$ is zero for $q < 0$. For any q it lives in cohomological degree $\leq q$. As a complex of Nisnevich sheaves it is quasi-isomorphic to $\mathbf{Z}$ for $q = 0$ and to $\mathbf{G}_m[-1]$ for $q = 1$.
2. $H^{p,p}(Spec(k)) = K_p^M(k)$ for any $p \geq 0$.
3. For any X in Sm/k one has

$$H^{p,q}(X) = CH^q(X, 2q - p)$$

where $CH^q(X, 2q - p)$ is the $(2q - p)$-th higher Chow group of cycles of codimension q.

4. In the etale topology, for n prime to the characteristic of k, the complex $\mathbf{Z}/n(q)$ is quasi-isomorphic to $\mu_n^{\otimes q}$, giving for the etale analog of $H^{p,q}$ the formula

$$H_{et}^{p,q}(X, \mathbf{Z}/n) := \mathbf{H}^p(X_{et}, \mathbf{Z}/n(q)) = H^p(X_{et}, \mu_n^{\otimes q}).$$

1.3 The category $SmCor(k)$ is the category with objects separated schemes smooth of finite type over k, for which a morphism $Z : X \to Y$ is a cycle $Z = \sum n_i Z_i$ on $X \times Y$ each of whose irreducible components Z_i is finite over X and projects onto a connected component of X. A morphism Z can be thought of as a finitely valued map from X to Y. For $x \in X$, with residue field $k(x)$, it defines a zero cycle $Z(x)$ on $Y_{k(x)}$, and the assumption made on Z implies that the degree of this 0-cycle is locally constant on X.

A morphism of schemes $f : X \to Y$ defines a morphism in $SmCor(k)$: the graph of f. This graph construction defines a faithful functor from Sm/k to the additive category $SmCor(k)$.

A *presheaf with transfers* is a contravariant additive functor from the category $SmCor(k)$ to the category of abelian groups. The embedding of Sm/k in

$SmCor(k)$ allows us to view a presheaf with transfers as a presheaf on Sm/k endowed with an extra structure. A sheaf with transfers (for a given topology on Sm/k, usually the Nisnevich topology) is a presheaf with transfers which, as a presheaf on Sm/k, is a sheaf. The Nisnevich and the etale topologies have the virtue that if F is a presheaf with transfers, the associated sheaf $a(F)$ carries a structure of a sheaf with transfers. This structure is uniquely determined by $F \to a(F)$ being a morphism of presheaves with transfers. For any sheaf with transfers G, one has

$$Hom(a(F), G) \xrightarrow{\sim} Hom(F, G)$$

(Hom of presheaves with transfers). All of this fails for the Zariski topology.

The complexes $\mathbf{Z}(q)$ (or $A(q)$) start life as complexes of sheaves with transfers.

1.4 A presheaf F on Sm/k is called *homotopy invariant* if $F(X) = F(X \times \mathbf{A}^1)$. As the point 0 of $\mathbf{A}^1$ defines a section of the projection of $X \times \mathbf{A}^1$ to X, for any presheaf of abelian groups F, $F(X)$ is naturally a direct factor of $F(X \times \mathbf{A}^1)$; it follows that the condition "homotopy invariant" is stable by kernels, cokernels and extensions of presheaves. The following construction is a derived version of the left adjoint to the inclusion

$$(homotopy \quad invariant \quad presheaves) \subset (all \quad presheaves)$$

(a) For S a finite set, let $A(S)$ be the affine space freely spanned (in the sense of barycentric calculus) by S. Over $\mathbf{C}$, or $\mathbf{R}$, $A(S)$ contains the standard topological simplex spanned by S. The schemes $\Delta^m := A(\{0, \ldots, n\})$ form a cosimplicial scheme.
(b) For F a presheaf, $C_\bullet(F)$ (the "singular complex of F") is the simplicial presheaf $C_n(F) : X \mapsto F(X \times \Delta^n)$.

Arguments imitated from topology show that for F a presheaf of abelian groups, the cohomology presheaves of the complex $C_*(F)$, obtained from $C_\bullet(F)$ by taking alternating sum of the face maps, are homotopy invariant. If F has transfers so do the $C_n(F)$ and hence the $H_n C_*(F)$. A basic theorem proved last year is:

Theorem 1. *Let F be a homotopy invariant presheaf with transfers over a perfect field with the associated Nisnevich sheaf $a_{Nis}(F)$. Then the presheaves with transfers*

$$X \mapsto H^i(X_{Nis}, a_{Nis}(F))$$

are homotopy invariant as well.

The particular case of this theorem for $i = 0$ claims the homotopy invariance of the sheaf with transfers $a_{Nis}(F)$.

Last year, the equivalence of a number of definitions of $\mathbf{Z}(q)$ was proven. Equivalence means: a construction of an isomorphism in a suitable derived category, implying an isomorphism for the corresponding motivic cohomology groups. For our present purpose the most convenient definition is as follows.

Let $\mathbf{Z}_{tr}(X)$ be the sheaf with transfers represented by X (on the category $SmCor(k)$). We set

$$K_q = \begin{cases} 0 & \text{for } q < 0 \\ \mathbf{Z}_{tr}(\mathbf{A}^q)/\mathbf{Z}_{tr}(\mathbf{A}^q - \{0\}) & \text{for } q \geq 0 \end{cases}$$

and $\mathbf{Z}(q) = C_*(K_q)[-q]$.

2.2 Motivic Homotopy Category

The motivic homotopy category $Ho_{\mathbf{A}^1,\bullet}(S)$ (pointed $\mathbf{A}^1$-homotopy category of S), for S a finite dimensional noetherian scheme, will be the category deduced from a category of simplicial sheaves by two successive localizations.[1]

One starts with the category Sm/S of schemes smooth over S, with the Nisnevich topology, and the category of pointed simplicial sheaves on Sm/S. For any site $\mathcal{S}$ (for instance $(Sm/S)_{Nis}$), there is a notion of *local equivalence* of (pointed) simplicial sheaves. It proceeds as follows:

(a) A sheaf G defines a simplicial sheaf G_* with all $G_n = G$ and all simplicial maps the identities. The functor $G \mapsto G_*$ has a left adjoint $F \mapsto \pi_0(F)$:

$$Hom(F_*, G_*) = Hom(\pi_0(F_*), G)$$

The sheaf $\pi_0(F_*)$ can be described as the equalizer of $F_1 \rightrightarrows F_0$, as well as the sheaf associated to the presheaf

$$U \mapsto \pi_0(|F_*(U)|)$$

The same holds in the pointed context. We will often write simply G for G_*.

(b) If F_* is a simplicial sheaf, and u a section of F_0 over U, one also disposes of sheaves $\pi_i(F_*, u)$ over U: the sheaves associated to the presheaves

$$V/U \mapsto \pi(|F_*(V)|, u).$$

(c) A morphism $F_* \to G_*$ is a local equivalence, if it induces an isomorphism on π_0 as well as, for any local section u of F_0, an isomorphism on all π_i. This applies also to pointed simplicial sheaves: one just forgets the marked point.

One defines $Ho_\bullet(Sm/S)$ as the category derived from the category of pointed simplicial sheaves on $(Sm/S)_{Nis}$ by formally inverting local equivalences. Until made more concrete, this definition could lead to set-theoretic difficulties, which we leave the reader to solve in its preferred way.

[1] In the Appendix we have assembled the properties of "localization" to be used in this talk and in the next.

For G a pointed sheaf on Sm/S, Proposition 7 applies to G_* and to the localization by local equivalences: one has

$$Hom_{Ho_\bullet}(F_*, G_*) = Hom(F_*, G_*) = Hom(\pi_0(F_*), G) \tag{0.1}$$

Definition 1. An object X of $Ho_\bullet(Sm/S)$ is called $\mathbf{A}^1$-local if for any simplicial sheaf Y, one has

$$Hom_{Ho_\bullet}(Y, X) \xrightarrow{\sim} Hom_{Ho_\bullet}(Y \times \mathbf{A}^1 / * \times \mathbf{A}^1, X)$$

At the right hand side, $/ * \times \mathbf{A}^1$ means that in the product, $* \times \mathbf{A}^1$ is contracted to a point, the new base point.

Proposition 1. *For G a pointed sheaf on Sm/S, the simplicial sheaf G_* is $\mathbf{A}^1$-local if and only if G is homotopy invariant.*

Proof. We have $\pi_0(Y \times \mathbf{A}^1) = \pi_0(Y) \times \mathbf{A}^1$, so that by (0.1) "$\mathbf{A}^1$-local" means that for any pointed sheaf Y, one has

$$Hom(Y, G) = Hom(Y \times \mathbf{A}^1 / * \times \mathbf{A}^1, G)$$

A morphism $Y \to G$ can be viewed as the data, for each $y \in Y(U)$, of $f(y) \in G(U)$, functorial in U and marked point going to marked point. A morphism $g : Y \times \mathbf{A}^1 \to G$ can similarly be described as data for $y \in Y(U)$ of $g(y) \in G(U \times \mathbf{A}^1)$. Homotopy invariance hence implies $\mathbf{A}^1$-locality. The converse is checked by taking for Y the disjoint sum of a representable sheaf and the base point. □

Definition 2. (1) A morphism $f : Y_1 \to Y_2$ in $Ho_\bullet(Sm/S)$ is an $\mathbf{A}^1$-equivalence if for any $\mathbf{A}^1$-local X, one has in $Ho_\bullet(Sm/S)$

$$Hom(Y_2, X) \xrightarrow{\sim} Hom(Y_1, X)$$

(2) The category $Ho_{\mathbf{A}^1,\bullet}(Sm/S)$ is deduced from $Ho_\bullet(Sm/S)$ by formally inverting $\mathbf{A}^1$-equivalences.

Remark 1. If a morphism in $Ho_\bullet(Sm/S)$ becomes an isomorphism in $Ho_{\bullet,\mathbf{A}^1}(Sm/S)$ it is an $\mathbf{A}^1$-equivalence. Indeed, if X in $Ho_\bullet(Sm/S)$ is $\mathbf{A}^1$-local, an application of Proposition 7 shows that for any Y,

$$Hom_{Ho_\bullet}(Y, X) \to Hom_{Ho_{\bullet,\mathbf{A}^1}}(Y, X)$$

is bijective. If $f : Y_1 \to Y_2$ in $Ho_\bullet(Sm/S)$ has an image in $Ho_{\bullet,\mathbf{A}^1}(Sm/S)$ which is an isomorphism, it follows that for any $\mathbf{A}^1$-local X, one has

$$Hom_{Ho_\bullet}(Y_2, X) \xrightarrow{\sim} Hom_{Ho_\bullet}(Y_1, X).$$

Such an f is an $\mathbf{A}^1$-equivalence.

Example 1. Arguments similar to those given before show that if G is a homotopy invariant pointed sheaf, then for any simplicial pointed sheaf F_*, one has

$$Hom_{Ho_{\mathbf{A}^1,\bullet}}(Sm/S)(F_*, G) = Hom(F_*, G) = Hom(\pi_0(F_*), G)$$

in particular, if U is smooth over S and if U_+ is the disjoint union of U and of a base point,

$$Hom_{Ho_{\mathbf{A}^1,\bullet}}(U_+, G) = G(U)$$

2.3 Derived Categories Vs. Homotopy Categories

For any topos T, which for us will be the category of sheaves on some site S, the pointed homotopy category $Ho_\bullet(S)$ as well as the derived category $D(S)$ are obtained by localization. For the derived category, one starts with the category of complexes of abelian sheaves. The subcategory of complexes living in homological degree ≥ 0 is naturally equivalent, by the Dold Puppe construction, to the category of simplicial sheaves of abelian groups. The equivalence is

$$N : (simplicial\ F_*) \mapsto complex(\bigcap_{i\neq 0} ker(\partial_i), \partial_0)$$

We will write K for the inverse equivalence. For S a point, and π an abelian group, $|K(\pi[n])|$ is indeed the Eilenberg–Maclane space $K(\pi, n)$. For a complex C not assumed to live in homological degree ≥ 0, we define

$$K(C) := K(\tau_{\geq 0}C)$$

where $\tau_{\geq n}C$ is the subcomplex in C of the form

$$\ldots C_{n+2} \stackrel{d_{n+1}}{\rightarrow} C_{n+1} \rightarrow ker(d_n) \rightarrow 0$$

Note that $C \mapsto \tau_{\geq 0}C$ is right adjoint to the inclusion functor

$$\text{(complexes in homological degree} \geq 0) \hookrightarrow \text{(all complexes)}$$

so that (N, K) form a pair of adjoint functors:

$$\text{(simplicial abelian sheaves)} \rightleftarrows \text{(complexes of sheaves)}$$

Theorem 2. *Assume that $S = Spec(k)$ with k perfect. Then, for F a presheaf with transfers, and U_+ as above, and $p \geq 0$*

$$Hom_{Ho_{\mathbf{A}^1,\bullet}}(U_+, K(F[p])) = \mathbf{H}^p(U_{Nis}, C_*(F))$$

In this theorem $K(F[p])$ is the simplicial sheaf of abelian groups whose normalized chain complex is F in homological degree p.

To prove the theorem we establish the chain of equalities,

$$\begin{aligned} &\mathbf{H}^p(U_{Nis}, C_*(F)) = Hom_{Ho_\bullet}(U_+, K(C_*(F)[p])) = \\ &= Hom_{Ho_{\bullet,\mathbf{A}^1}}(U_+, K(C_*(F)[p])) = Hom_{Ho_{\bullet,\mathbf{A}^1}}(U_+, K(F[p])), \end{aligned} \tag{1.1}$$

the first equality is proved right before Proposition 4, the second right after Proposition 5 and the last one follows from Lemma 1.

Let $forget$ be the forgetting functor from abelian sheaves to sheaves of sets. Its left adjoint is $F \mapsto \mathbf{Z}[F]$: the sheaf associated to the presheaf

$$U \mapsto \text{(abelian group freely generated by } F(U))$$

In the pointed context , the adjoint is

$$(F, *) \mapsto \tilde{\mathbf{Z}}(F) : \mathbf{Z}(F)/\mathbf{Z}(*)$$

We have the same adjunction for (pointed) simplicial objects.

Proposition 2. *On a site with enough points (and presumably always), one has*

(1) The functor $F_* \mapsto N\mathbf{Z}(F_*)$ from pointed simplicial sheaves to complexes of abelian groups transforms local equivalences into quasi-isomorphisms
(2) The right adjoint $C \mapsto forget(K(C))$ transforms quasi-isomorphisms to local equivalences.

The assumption "enough points" applies to Sm/k with the Nisnevich topology: for any U in Sm/k and any point x of U, $F \mapsto F(Spec(\mathcal{O}^h_{U,x}))$ is a point, and they form a conservative system.

Proof. Local equivalence (resp. quasi-isomorphism) can be checked point by point, and the two functors considered commute with passage to the fiber at a point. This reduces our proposition to the case when S is just a point, i.e. to usual homotopy theory. In that case, (1) boils down to the fact that a weak equivalence induces an isomorphism on reduced homology, a theorem of Whitehead, and (1.1) reduces to the fact: for a complex C, $\pi_i(K(C))$, computed using any base point, is $H_i(C)$. The $\pi_i(K(C))$ are easy to compute because $K(C)$ has the Kan property. $\square$

Applying Proposition 8, we deduce from Proposition 2 the following.

Proposition 3. *Under the same assumptions as in Proposition 2, for F_* a pointed simplicial sheaf and C a complex of abelian sheaves, one has*

$$Hom_{Ho_\bullet}(F_*, K(C)) = Hom_D(N\tilde{\mathbf{Z}}(F_*), C) \tag{3.1}$$

Let F be a sheaf and F_+ be deduced from F by adjunction of a base point. We also write F and F_+ for the corresponding "constant" simplicial sheaf. One has

$$N\tilde{\mathbf{Z}}(F_+) = N\mathbf{Z}(F) = (\mathbf{Z}(F) \text{ in degree zero})$$

For the pointed simplicial sheaf F_+, the group $Hom_D(\mathbf{Z}(F), C)$ which now occurs at the right-hand side of (3.1) can be interpreted as hypercohomology of C "over F viewed as a space", i.e. in the topos of sheaves over F. For F defined by an object U of the site S, this is the same as hypercohomology of the site S/U. As we do not want to assume C bounded below (in cohomological numbering), checking this requires a little care.

For a complex of sheaves K over a site S, not necessarily bounded below, $\mathbf{H}^0(S, K)$ can be defined as the Hom group in the derived category $Hom_D(\mathbf{Z}, K)$. For F in a topos T and the topos T/F: "F viewed as a space", besides the morphism of toposes $(T/F) \xrightarrow{j} T$, i.e. the adjoint pair (j^*, j_*), we have for abelian sheaves an adjoint pair $(j_!, j^*)$, with $j_!$ and j^* both exact. By Proposition 8, $(j_!, j^*)$ induce an adjoint pair for the corresponding derived categories. As $j_!\mathbf{Z} = \mathbf{Z}[F]$, we get

$$Hom_D(\mathbf{Z}(F), C) = \mathbf{H}^0(T/F, j^*(C)) \tag{3.2}$$

hence

$$Hom_{Ho_\bullet}(F_+, K(C)) = \mathbf{H}^0(T/F, j^*(C)) \tag{3.3}$$

Let us consider the particular case of Sm/k with the Nisnevich topology. For any complex of sheaves, (3.3) gives for U smooth over k

$$Hom_{Ho_\bullet}(U_+, K(C)) = \mathbf{H}^0(U_{big-Nis}, C) \tag{3.4}$$

Here, $U_{big-Nis}$ is the site $(Sm/S)/U$ with the Nisnevich topology. It has however the same hypercohomology as the small Nisnevich site U_{Nis}. Indeed, one has a morphism $\epsilon : U_{big,Nis} \to U_{Nis}$ and the functors ϵ^* and ϵ_* are exact. One again applies Proposition 8. If we apply (3.4) to a translate (shift) of C, we get

$$Hom_{Ho_\bullet}(U_+, K(C[p])) = \mathbf{H}^p(U_{Nis}, C) \tag{3.5}$$

Applying (3.5) to $C_*(F)$ we get the first equality in (1.1).

Proposition 4. *Let C be a complex of abelian sheaves on Sm/k. The following conditions are equivalent:*

1. *$K(C)$ is $\mathbf{A}^1$-local.*
2. *For $i \leq 0$, the functor $U \mapsto \mathbf{H}^i(U, C)$ is homotopy invariant.*
3. *For any complex L in cohomological degree ≤ 0, one has in the derived category*

$$Hom(L \otimes \mathbf{Z}(\mathbf{A}^1), C) \xrightarrow{\sim} Hom(L, C).$$

Proof. By Proposition 3, condition (1) can be rewritten: for any pointed simplicial sheaf F_*.

$$Hom_D(N\tilde{\mathbf{Z}}(F_*), C) = Hom_D(N\tilde{\mathbf{Z}}(F_* \times \mathbf{A}^1 / * \times \mathbf{A}^1))$$

The operation $F_* \mapsto F_* \times \mathbf{A}^1 / * \times \mathbf{A}^1$ is better written as a smash product $F_* \wedge \mathbf{A}^1_+$ with $\mathbf{A}^1_+$. For pointed sets E and F, $\tilde{\mathbf{Z}}(E \wedge F) = \tilde{\mathbf{Z}}(E) \otimes \tilde{\mathbf{Z}}(F)$. It follows that

$$\tilde{\mathbf{Z}}(F_* \times \mathbf{A}^1 / * \times \mathbf{A}^1) = \tilde{\mathbf{Z}}(F_* \wedge \mathbf{A}^1_+) = \tilde{\mathbf{Z}}(F_*) \otimes \tilde{\mathbf{Z}}(\mathbf{A}^1_+) = \tilde{\mathbf{Z}}(F_*) \otimes \mathbf{Z}(\mathbf{A}^1)$$

(isomorphisms of simplicial sheaves), hence

$$N\tilde{\mathbf{Z}}(F_* \times \mathbf{A}^1 / * \times \mathbf{A}^1) = N\tilde{\mathbf{Z}}(F_*) \otimes \mathbf{Z}(\mathbf{A}^1)$$

It follows that (1) is the particular case of (3) for L of the form $\tilde{\mathbf{Z}}(F_*)$. Similarly, (2) is the particular case of (3) for L of the form $\mathbf{Z}(U)[i]$, with $i \geq 0$.

The suspension $\Sigma^i F_*$ of a simplicial pointed sheaf F_* is its smash product with the simplicial sphere S^i (the i-simplex modulo its boundary). It follows that

$$\tilde{\mathbf{Z}}(\Sigma^i F_*) = \tilde{\mathbf{Z}}(F_*) \otimes \tilde{\mathbf{Z}}(S^i)$$

(isomorphism of simplicial sheaves), and by Eilenberg–Zilber, the normalized complex $N\mathbf{Z}(\Sigma^i F_*)$ is homotopic to the tensor product of the normalized complexes of $\tilde{\mathbf{Z}}(F_*)$ and $\tilde{\mathbf{Z}}(S^i)$. The latter is simply $\mathbf{Z}[i]$:

$$N\tilde{\mathbf{Z}}(\Sigma^i F_*) \cong \tilde{\mathbf{Z}}(F_*)[i]$$

This is just a high-brown way of telling that the reduced homology of a suspension is just a shift of the reduced homology of the space one started with.

Applying this to $F_* = U_+$, one obtains that (1)⇒(3). Indeed, $\tilde{\mathbf{Z}}(\Sigma^i U_+)$ is homotopic to $\tilde{\mathbf{Z}}(U_+)[i] = \mathbf{Z}(U)[i]$.

We now prove that (2)⇒(1). For L a complex, let (*) be the statement that the conclusion of (3) holds for all $L[i]$, $i \geq 0$. The assumption (2) is that (*) holds for L reduced to $\mathbf{Z}(U)$ in degree 0, and we will conclude that it holds for all L in cohomological degree ≤ 0 by "devissage":

(a) The case of a sum of $\mathbf{Z}(U)$, in degree zero, follows from Corollary 10.
(b) Suppose that L is bounded, is in cohomological degree ≤ 0 and that (*) holds for all L^n.

The functors

$$h' : L \mapsto Hom^n(L, C) \tag{3.6}$$
$$h'' : L \mapsto Hom^n(L \otimes \mathbf{Z}(\mathbf{A}^1), C)$$

are contravariant cohomological functors, hence give rise to convergent spectral sequences

$$E_1^{pq} = h^q(L^{-p}) \Rightarrow h^{p+q}(L).$$

One has a morphism of spectral sequences

$$E(\text{for } h') \to E(\text{for } h'')$$

which is an isomorphism for $q \leq 0$, and both E^{pq} vanish for $p < 0$ or p large. It follows that $h'^n(L) \xrightarrow{\sim} h''^n(L)$ for $n \leq 0$, i.e. that L satisfies (*).

The same argument can be expressed as an induction on the number of i such that $L^i \neq 0$. If n is the largest (with $n \leq 0$), the induction assumption applies to $\sigma^{<n}L$, even to $(\sigma^{<n}L)[-1]$, and one concludes by the long exact sequence defined by

$$0 \to L^n[-n] \to L \to \sigma^{<n}L \to 0$$

(c) Expressing L as the inductive limit of the $\sigma^{\geq -n}L$ and using Proposition 11, one sees that we need not assume that L is bounded.
(d) If $L' \to L''$ is a quasi-isomorphism, $L' \otimes \mathbf{Z}(\mathbf{A}^1) \to L'' \otimes \mathbf{Z}(\mathbf{A}^1)$ is one too (flatness of $\mathbf{Z}(\mathbf{A}^1)$), and (*) holds for L' if and only if it holds for L''.
(e) Any abelian sheaf L is a quotient of a direct sum of sheaves $\mathbf{Z}(U)$. For instance, the sum over (U, s), $s \in \Gamma(U, L)$, of $\mathbf{Z}(U)$ mapping to L by s. It follows that L admits a resolution L^* by such sheaves. By (a) and (c), L^* satisfies (*). It follows from (d) that L satisfies (*) and then by (c) that any complex in degree ≤ 0 satisfies (*). □

2.4 Application to Presheaves with Transfers

Let F be a presheaf with transfers. A formal argument [7] shows that the presheaves with transfers $H^iC_*(F)$ are homotopy invariant. By the basic result (Theorem 1) recalled in the first lecture, it follows that for any U, one has

$$\mathbf{H}^*(U, C_*(F)) = \mathbf{H}^*(U \times \mathbf{A}^1, C_*(F)) \tag{4.1}$$

Indeed, as U and $U \times \mathbf{A}^1$ are of finite cohomological dimension, both sides are abutment of a convergent spectral sequence

$$E_2^{pq} = H^p(U, H^qC_*(F)) \Rightarrow \mathbf{H}^{p+q}(U, C_*(F))$$

and the same for $U \times \mathbf{A}^1$. By Theorem 1 applied to $H^qC_*(F)$,

$$H^p(X, H^qC_*(F)) := H^p(X, aH^qC_*(F))$$

is the same for $X = U$ or for $X = U \times \mathbf{A}^1$. Applying (3.5), we conclude from Proposition 4((2)⇒(1)) that

Proposition 5. *For k perfect, if F is a presheaf with transfers, for all p, $K(C_*(F)[p])$ is $\mathbf{A}^1$-local.*

Combining Proposition 5 with Proposition 7 we get the second equality in (1.1).

2.5 End of the Proof of Theorem 2

For any pointed simplicial sheaf $G_\bullet$, $C_\bullet(G_\bullet)$ is a pointed bisimplicial sheaf of which one can take the diagonal $\Delta C_\bullet(G_\bullet)$. For any pointed sheaf G, one has a natural map $G \to C_\bullet(G)$, and for a pointed simplicial sheaf $G_\bullet$, those maps for the G_n induce

$$a : G_\bullet \to \Delta C_\bullet(G_\bullet)$$

Proposition 6. *The morphism $a : G_\bullet \to \Delta C_\bullet(G_\bullet)$ is an $\mathbf{A}^1$-equivalence.*

Proof. We deduce Proposition 6 from Proposition 20.

The two maps $0, 1 : F_\bullet \to F_\bullet \wedge \mathbf{A}^1_+$ are equalized by $F_\bullet \wedge \mathbf{A}^1_+ \to F_\bullet$, hence become equal in the $\mathbf{A}^1$-homotopy category. If two maps of pointed simplicial sheaves $F_\bullet \rightrightarrows G_\bullet$ factor as $F_\bullet \rightrightarrows F_\bullet \wedge \mathbf{A}^1_+ \to G_\bullet$, they also become equal. By the adjunction of $\wedge \mathbf{A}^1_+$ and of $C_1(-) = \underline{Hom}(\mathbf{A}^1_+, -)$, such a factorization can be rewritten as

$$F_\bullet \to C_1(G_\bullet) \to G_\bullet$$

Particular case: the maps $C_1(G_\bullet) \to G_\bullet$, become equal in the homotopy category. Evaluated on X, these maps are the restriction maps $0^*, 1^* : G_\bullet(X \times \mathbf{A}^1) \to G_\bullet(X)$.

The affine space $\mathbf{A}^n$ is homotopic to a point in the sense that $H : \mathbf{A}^1 \times \mathbf{A}^n \to \mathbf{A}^n : (t, x) \mapsto tx$ interpolates between the identity map (for $t = 1$) and the constant map 0 (for $t = 0$). The map H induces

$$G_\bullet(S \times \mathbf{A}^n) \to G_\bullet(S \times \mathbf{A}^n \times \mathbf{A}^1)$$

and, composing with 0, 1 in $\mathbf{A}^1$, we obtain that

$$G_\bullet(S \times \mathbf{A}^n) \rightrightarrows G_\bullet(S \times \mathbf{A}^n)$$

the identity map, and the map induced by $0 : \mathbf{A}^n \to \mathbf{A}^n$, are equal in the $\mathbf{A}^1$-homotopy category. The map of simplicial sheaves $G_\bullet \to C_n G_\bullet$ is hence an $\mathbf{A}^1$-equivalence. It has as inverse in the $\mathbf{A}^1$-homotopy category the map induced by

$0 : Spec(k) \to \mathbf{A}^n$ and one applies Remark 1. We now apply Proposition 20 to the bisimplicial sheaves

$$G_{pq} := G_p$$
$$H_{pq} := C_q G_p : S \mapsto G_p(S \times \Delta^q)$$

and to the natural map $G_{pq} \to H_{pq}$. For fixed q, this is just $G(S) \to G(S \times \mathbf{A}^q)$, and Proposition 20 gives Proposition 6. □

To prove the last equality in (1.1), it suffices to show that:

Lemma 1. *For any abelian sheaf F, $F[p] \to C_*(F)[p]$ induces an $\mathbf{A}^1$-equivalence from $K(F[p])$ to $K(C_*(F)[p])$.*

Proof. For G a monoid (with unit), the pointed simplicial set $B_\bullet G$ is given by

$$B_n G = \left\{ \begin{array}{c} \text{functors from the ordered set } (0, \ldots, n) \text{ viewed as a category} \\ \text{to } G \text{ viewed as a category with one object} \end{array} \right\}$$

This construction can be sheafified, and can be applied termwise to a simplicial sheaf of monoids, leading to a pointed bisimplicial sheaf of which one can take the diagonal

$$BG_\bullet := \Delta B_\bullet(G_\bullet)$$

This construction commutes with the construction $G_\bullet \to \Delta C_\bullet(G_\bullet)$. Indeed, $B_n G_p$ is naturally isomorphic to G_p^n, the operation C_m commutes with products, and $B(\Delta C_\bullet(G_\bullet))$ and $\Delta C_\bullet(BG_\bullet)$ are both diagonals of the trisimplicial pointed sheaf $C_\bullet B_\bullet G_\bullet$.

For abelian simplicial sheaves, the operation B gives again abelian simplicial sheaves, hence can be iterated, and $\Delta C_\bullet$ commutes with B^n.

Via Dold–Puppe construction, B corresponds, up to homotopy, to the shift [1] of complexes:

$$NBG_\bullet \cong (NG_\bullet)[1].$$

This can be viewed as an application of the Eilenberg–Zilber Theorem (see [9, Theorem 8.5.1]): one has

$$NBG_\bullet \cong BG_* \cong Tot B_* G_* \quad \text{(Eilenberg–Zilber)},$$

and for each G_q, the normalization of $B_\bullet G_q$ is just $G_q[1]$, so that the double complexes $B_* G_*$ and $H_{pq} := G_q$ for $p = 1$, 0 otherwise, have homotopic Tot.

If $G_\bullet$ is an abelian simplicial sheaf, applying Proposition 6 to $B^p G_\bullet$, we obtain that

$$B^p G_\bullet \to \Delta C_\bullet B^p G_\bullet = B^p \Delta C_\bullet G_\bullet \qquad (6.1)$$

is an $\mathbf{A}^1$-equivalence. The functor K transforms chain homotopy equivalences into simplicial equivalences. For any simplicial abelian group $L_\bullet$ (to be $G_\bullet$ or $\Delta C_\bullet G_\bullet$), we hence have a simplicial homotopy equivalence

$$B^p L_\bullet = KNB^p L_\bullet \cong K((NL_\bullet)[p])$$

Simplicial homotopy equivalences being $\mathbf{A}^1$-equivalences, we conclude that (6.1) induces an $\mathbf{A}^1$-equivalence

$$K((NG_\bullet)[p]) \to K(N(\Delta C_\bullet G_\bullet)[p])$$

□

2.6 Appendix: Localization

Let C be a category and S be a set of morphisms of C. The localized category $C[S^{-1}]$ is deduced from C by "formally inverting all $s \in S$". With this definition, it is clear that one has a natural functor $loc : C \to C[S^{-1}]$, bijective on the set of objects, and that for any category D,

$$F \mapsto F \circ loc : Hom(C[S^{-1}], D) \to Hom(C, D)$$

is a bijection from $Hom(C[S^{-1}], D)$ to the set of functors from C to D transforming morphisms in S into isomorphisms.

If one remembers that the categories form a 2-category, and if one agree with the principle that one should not try to define a category more precisely than up to equivalence (unique up to unique isomorphism), the universal property of $C[S^{-1}]$ given above is doubly unsatisfactory. The easily checked and useful universal property is the following: $F \mapsto F \circ loc$ is an equivalence from the category $Hom(C[S^{-1}], D)$ to the full subcategory of $Hom(C, D)$ consisting of the functors F which map S to isomorphisms.

Proposition 7. *If Y in C is such that the functor*

$$h_Y : C^{op} \to Sets : X \mapsto Hom_C(X, Y)$$

transforms maps in S into bijections, then

$$Hom_C(X, Y) \xrightarrow{\sim} Hom_{C[S^{-1}]}(X, Y)$$

Proof. By Yoneda construction $Y \mapsto h_Y$, C embeds into the category $C^\wedge$ of contravariant functors from C to *Sets*, while $C[S^{-1}]$ embeds into $C[S^{-1}]^\wedge$, identified by (a) with the full subcategory of $C^\wedge$ consisting of F transforming S into bijections. For Y in C, with image $\bar{Y}$ in $C[S^{-1}]$, and for any F in $C(S^{-1})^\wedge \subset C^\wedge$, one has in $C^\wedge$

$$Hom(h_Y, F) = Hom(h_{\bar{Y}}, F).$$

Indeed, by (a) and Yoneda lemma for C and $C[S^{-1}]$ both sides are $F(Y)$. This means that $h_{\bar{Y}}$ is the solution of the universal problem of mapping h_Y into an object of $C[S^{-1}]^\wedge \subset C^\wedge$. In particular, for Y as in (b), i.e. in $C[S^{-1}]^\wedge$, $h_{\bar{Y}}$ coincides with h_Y, as claimed by (b). □

Proposition 8. *Let (L, R) be a pair of adjoint functors between categories C and D. Let S and T be sets of morphisms in C and D. Assume that F maps S to T and that G maps T to S. Then the functors $\bar{L}$, $\bar{R}$ between $C[S^{-1}]$ and $D[T^{-1}]$ induced by L and R again form an adjoint pair.*

Proof. The functors $\bar{L}$ and $\bar{R}$ induced by F and G are characterized by commutative diagrams

$$\begin{array}{ccc} C & \xrightarrow{L} & D \\ \downarrow & & \downarrow \\ C[S^{-1}] & \xrightarrow{\bar{L}} & D[T^{-1}] \end{array} \qquad \begin{array}{ccc} D & \xrightarrow{R} & C \\ \downarrow & & \downarrow \\ D[T^{-1}] & \xrightarrow{\bar{R}} & C[S^{-1}] \end{array}$$

Adjunction can be expressed by the data of $\epsilon : Id \to RL$ and $\eta : LR \to Id$ such that the compositions

$$R \to RLR \to R$$

$$L \to LRL \to L$$

are the identity automorphisms of R and L respectively (see, e.g. [6]).

By the universal property of localization, ϵ induces a morphism $\bar{\epsilon} \to \bar{R}\bar{L}$, indeed, to define such a morphism amounts to defining a morphism $loc \to \bar{R}\bar{L}loc$, and $\bar{R}\bar{L}loc = locRL$. Similarly, η induces $\bar{\eta} : \bar{L}\bar{R} \to Id$. The morphism $\bar{L} \to \bar{L}\bar{R}\bar{L} \to \bar{L}$ is induced by $L \to LRL \to L$, similarly for $\bar{R} \to \bar{R}\bar{L}\bar{R} \to \bar{R}$, and the proposition follows. □

Proposition 9. *Suppose that:*

1. *The localization $C[S^{-1}]$ gives rise to a right calculus of fractions.*
2. *Coproducts exist in C, and S is stable by coproducts.*

Then, a coproduct in C is also a coproduct in $C[S^{-1}]$.

For the definition of "gives rise to a right calculus of fractions" see [10]. It implies that for X in C, the category of $s : X' \to X$ with s in S is filtering, and that

$$Hom_{C[S^{-1}]}(X, Y) = colim_{s:X' \to X} Hom_C(X', Y)$$

Proof. For X in C, let (S/X) be the filtering category of morphisms $X' \to X$ in S. For X the coproduct of X_α, $\alpha \in A$, one has a functor "coproduct":

$$\prod(S/X_\alpha) \to (S/X)$$

It is cofinal: for $s : X' \to X$ in S, one can construct a diagram

$$\begin{tikzcd} X'_\alpha \arrow[r] \arrow[d] & X_\alpha \arrow[d] \\ X' \arrow[r] & X \end{tikzcd}$$

with $s_\alpha : X'_\alpha \to X$ in S, and $\coprod s_\alpha$ dominates s. For any Y, it follows that

$$Hom_{C[S^{-1}]}(X, Y) = colim_{(S/X)} Hom_C(X', Y) =$$

$$= colim_{\prod(S/X_\alpha)} Hom(\coprod X'_\alpha, Y) = colim_{\prod(S/X_\alpha)} \prod Hom(X'_\alpha, Y) =$$

$$= \prod colim_{S/X_\alpha} Hom(X'_\alpha, Y) = \prod Hom_{C[S^{-1}]}(X_\alpha, Y),$$

meaning that X is also the coproduct of the X_α in $C[S^{-1}]$. □

Corollary 10. *Suppose that in the abelian category A arbitrary direct sums exist and are exact. Then, arbitrary direct sums exist in the derived category $D(A)$, and the localization functor*

$$C(A) \to D(A)$$

commutes with direct sums.

Proof. The functor $C(A) \to D(A)$ factors through the category $K(A)$ of complexes and maps up to homotopy. Direct sums in $C(A)$ are also direct sums in $K(A)$. Indeed,

$$Hom_{K(A)}(\oplus K_\alpha, L) = H^0 Hom^\bullet(\oplus K_\alpha, L) =$$

$$= H^0 \prod Hom^\bullet(K_\alpha, L) = \prod H^0 Hom^\bullet(K_\alpha, L),$$

as $\prod$ is exact for abelian groups. Exactness of $\oplus$ in A ensures that a direct sum of quasi-isomorphisms is again a quasi-isomorphism, and Proposition 9 applies to $K(A)$ and the set S of quasi-isomorphisms, proving the corollary. □

If A_i, $i \geq 0$ is an inductive system of objects of A, the colimit of A_i is the cokernel

$$\oplus A_i \xrightarrow{d} \oplus A_i \to colim\, A_i \to 0$$

of the difference of the identity map and of the sum of the $A_i \to A_{i+1}$. If taking the inductive limit of a sequence is an exact functor, the map d is injective: it is the colimit of the

$$\oplus_{i=0}^{n} A_i \to \oplus_{i=0}^{n+1} A_i$$

each of which is injective, as its graded for the filtration by the $\oplus_{i \geq p} A_i$ is the identity inclusion.

Under the assumptions of Corollary 10 if a complex K is the colimit of an inductive sequence $K_{(i)}$, and if the sequence

$$0 \to \oplus K_{(i)} \xrightarrow{d} \oplus K_{(i)} \to K \to 0 \tag{10.1}$$

is exact, then for any L, the long exact sequence of cohomology reads

$$\to Hom(K, L) \to \prod Hom(K_{(i)}, L) \xrightarrow{d} \prod Hom(K_{(i)}, L) \to$$

The kernel of d is simply the projective limit of the $Hom(K_{(i)}, L)$. The cokernel is lim^1. One concludes.

Proposition 11. *Suppose that in A countable direct sums exist and are exact. If the complex K is the colimit of the $K_{(i)}$, and if the sequence (10.1) is exact, for instance if either:*

1. *In A inductive limits of sequences are exact.*
2. *In each degree n, each $K^n_{(i)} \to K^n_{(i+1)}$, is the inclusion of a direct factor.*

then, one has a short exact sequence

$$0 \to lim^1 Hom(K_{(i)}, L[-1]) \to Hom(K, L) \to lim Hom(K_{(i)}, L) \to 0$$

Proof. It remains to check that condition (2) implies the exactness of (10.1). This is to be seen degree by degree. By assumption, the $A_i := K^n_{(i)}$, have decompositions compatible with the transition maps $A_i = \oplus_{j=0}^{i} B_i$. A corresponding decomposition of (10.1) in direct sum follows, and we are reduced to check exactness of the particular case $(10.1)_{(A,n)}$ of (10.1) when $B_i = 0$ for $i \neq n$, i.e. when A_i is a fixed A from $i \geq n$, and is 0 otherwise. Let $(10.1)_0$ be the sequence $(10.1)_{\mathbf{Z},n}$ in Ab for $A = \mathbf{Z}$. It is a split exact sequence of free abelian groups. Because direct sums exist, $L \otimes A$, for L a free abelian group is defined and functorial in L. It is a sum of copies of A, indexed by a basis of L, and is characterized by

$$Hom(L \otimes A, B) = Hom(L, Hom(A, B))$$

(functorial in B). The sequence $(10.1)_{A,n}$ is $(10.1)_0 \otimes A$ and, $(10.1)_0$ being split exact, it splits and in particular is exact. $\square$

The truncation $\sigma_{\leq n} K = \sigma^{\geq -n} K$ of a complex K is the subcomplex which coincides with K in homological degree $\leq n$ and is 0 in homological degree $> n$. For any complex K, one has

$$K = colim \sigma_{\leq n} K$$

and this colimit satisfies condition (2) of Proposition 11. It follows that

Corollary 12. *Under the assumptions of Corollary 10, for any K and L, one has a short exact sequence*

$$0 \to lim^1 Hom(\sigma_{\leq n} K, L[-1]) \to Hom(K, L) \to lim Hom(\sigma_{\leq n} K, L) \to 0$$

3 $\mathbf{A}^1$-Equivalences of Simplicial Sheaves on G-Schemes

3.1 Sheaves on a Site of G-Schemes

We fix a base scheme S, supposed to be separated noetherian and of finite dimension; fiber product $X \times_S Y$ will be written simply as $X \times Y$. We also fix a group scheme G over S, supposed to be finite and flat. We note h_X the representable sheaf defined by X.

Let QP/G be the category of schemes quasi-projective over S, given with an action of G. Any X in QP/G admits an open covering (U_i) by affine open subschemes which are G-stable. This makes it possible to define a reasonable quotient X/G in the category of schemes over S (rather than in the larger category of algebraic spaces). For each U_i, U_i/G is defined as the spectrum of the equalizer

$$\mathcal{O}(U_i) \rightrightarrows \mathcal{O}(U_i \times G),$$

and X/G is obtained by gluing the U_i/G. It is a categorical quotient, i.e. the coequalizer of $G \times X \rightrightarrows X$. The map $X \to X/G$ is finite, open, and the topological space $|X/G|$ is the coequalizer of the map of topological spaces $|G \times X| \rightrightarrows |X|$. One can show that X/G is again quasi-projective. Remark 2 below shows that this fact, while convenient for the exposition, is irrelevant.

One defines on QP/G a pretopology [3, II.1.3] by taking as coverings the family of etale maps $Y_i \to X$ with the following property: X admits a filtration by closed equivariant subschemes $\emptyset = X_n \subset \cdots \subset X_1 \subset X_0 = X$ such that for each j, some map $Y_i \to X$ has a section over $X_j - X_{j+1}$. The *Nisnevich topology* on QP/G is the topology generated by this pretopology. The category QP/G with the Nisnevich topology is the *Nisnevich site* $(QP/G)_{Nis}$.

Remark 1. The corresponding topos is not the classifying topos of [2, IV.2.5]. A morphism $X \to Y$ can become a Nisnevich covering after forgetting the action of G, and not be a Nisnevich covering. Example: $S = Spec(k)$, $G = \mathbf{Z}/2$, $X = S$, $Y = S \coprod S$ and G permutes two copies of S in Y.

Remark 2. Let $(affine/G)_{Nis}$ be the site defined as above, with "quasi-projective" replaced by "affine". It is equivalent to $(QP/G)_{Nis}$, in the sense that restriction to $(affine/G)_{Nis}$ is an equivalence from the category of sheaves on $(QP/G)_{Nis}$ to the category of sheaves on $(affine/G)_{Nis}$.

Remark 3. If G is the trivial group e, the definition given above recovers the usual Nisnevich topology. For $G = e$, the condition usually considered: "every point x of X is the image of a point with the same residue field of some Y_i", is indeed equivalent to the condition imposed above. This is checked by noetherian induction: if a generic point ξ of X can be lifted to Y_i, some open neighborhood $U \subset X$ of ξ can be lifted to Y_i, and one applies the induction hypothesis to $X_1 = (X - U)_{red}$.

We write $(QP)_{Nis}$ for the category of quasi-projective schemes over S, with the Nisnevich topology.

Lemma 2. *If $\mathcal{U} : Y_i \to X$ $(i \in I)$ is a covering of X in $(QP/G)_{Nis}$, there is a covering $\mathcal{V}$ of X/G in $(QP)_{Nis}$ whose pull-back to X is finer than $\mathcal{U}$.*

Proof. Fix a filtration $\emptyset = X_n \subset \cdots \subset X_0 = X$ as in the definition of the Nisnevich topology. We write q for the quotient map $X \to X/G$. For x in X/G, $(q^{-1}(x))_{red}$ is in some $X_j - X_{j+1}$, by equivariance of the X_j, and one of the maps $Y_i \to X$ has an equivariant section s over $X_j - X_{j+1}$. Let $(X/G)^h_x$ be the henselization of X/G at x. The map q being finite, the pull-back of $(X/G)^h_x$ to X is the coproduct of the X^h_y for $q(y) = x$. The map from Y_i to X being etale, the section s, restricted to $(q^{-1}(y))_{red}$, extends uniquely to a section (automatically equivariant) of Y_i over $\coprod_{q(y)=x} X^h_y$. Writing $(X/G)^h_x$ as the limit of etale neighborhoods of x, one finds that x has an etale neighborhood $V(x)$ such that Y_i has an equivariant section over $X \times_{X/G} V(x)$. The $V(x)$ form the required covering $\mathcal{V}$. $\square$

We define the G-local henselian schemes to be the schemes Y obtained in the following way. For X in (QP/G), y a point of X/G, and $(X/G)^h_y$ the henselization of X/G at y, take the fiber product $Y := X \times_{X/G} (X/G)^h_y$. As X is finite over X/G, this fiber product is a finite disjoint union of local henselian schemes, and G-local henselian schemes are simply the G-equivariant finite disjoint unions of Y of local henselian schemes, for which Y/G is local.

Proposition 13. *If Y is G-local henselian, the functor $X \mapsto Hom(Y, X)$ is a point of the site $(QP/G)_{Nis}$, i.e. it defines a morphism of the punctual site $(Sets)$ to $(QP/G)_{Nis}$. If $Y = X \times_{X/G} (X/G)^h_y$, the corresponding fiber functor is $F \mapsto colim F(X \times_{X/G} V)$, the colimit being taken over the etale neighborhoods of y in X/G. The collection of fiber functors so obtained is conservative.*

Proof. The functor $X \mapsto Hom(Y, X)$ commutes with finite limits. It follows from Lemma 2 that it transforms coverings into surjective families of maps, hence is a morphism of sites $(Sets) \to (QP/G)_{Nis}$.

To check that the resulting set of fiber functors is conservative, it suffices to check that a family of etale $f_i : U_i \to X$ is a covering if for any G-local henselian Y,

$$\coprod Hom(Y, U_i) \to Hom(Y, X)$$

is onto. The proof, parallel to that of Lemma 2 is left to the reader. □

3.2 The Brown–Gersten Closed Model Structure on Simplicial Sheaves on G-Schemes

We recall that a commutative square of simplicial sets (or pointed simplicial sets)

$$\begin{array}{ccc} K & \longrightarrow & L \\ \downarrow & & \downarrow \\ M & \longrightarrow & N \end{array} \tag{13.1}$$

is *homotopy cartesian* (or a *homotopy pull-back square*) if, when L is replaced by L' weakly equivalent to it and mapping to N by (Kan) fibration: $L \xrightarrow{\cong} L' \to N$, the map from K to $L' \times_N M$ is a weak equivalence.

Definition 3. A simplicial presheaf $F_\bullet$ on $(QP/G)_{Nis}$ is *flasque* if $F(\emptyset)$ is contractible and if for any (upper) distinguished square:

$$\begin{array}{ccc} B & \longrightarrow & Y \\ \downarrow & & \downarrow p \\ A & \xrightarrow{j} & X \end{array}$$

(p – etale, j open embedding, $B = p^{-1}(A)$ and $Y - B \cong X - A$), the square

$$\begin{array}{ccc} F(X) & \longrightarrow & F(Y) \\ \downarrow & & \downarrow p \\ F(A) & \xrightarrow{j} & F(B) \end{array}$$

is homotopy cartesian.

Theorem 3. *Let $f : F_\bullet \to F'_\bullet$ be a morphism of flasque simplicial presheaves. If the induced morphism of simplicial sheaves $aF_\bullet \to aF'_\bullet$ is a local equivalence, then, for any U in QP/G, $F_\bullet(U) \to F'_\bullet(U)$ is a weak equivalence.*